

Dynamics of a two-photon laser

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The high degree of temporal coherence of laser light arises from a complex interplay between the fundamental light-matter interactions of absorption, spontaneous emission, and stimulated emission. The coherence properties of the generated light can be altered significantly, and often in a surprising manner, by modifying the type of light-matter interaction on which the laser is based. As an example, the two-photon laser [1, 2] is based on the higher-order two-photon stimulated emission process, whereby two incident photons stimulate an excited atom to a lower energy state and four photons are scattered.

We have recently achieved degenerate two-photon optical lasing [3] using laser-driven potassium atoms [4] contained in a very high finesse optical resonator. This new type of quantum oscillator displays the expected turn-on behavior whereby it can only turn on when an injected coherent field brings the intercavity light above a critical value. Once the laser is above threshold, we observe polarization instabilities and chaos even though the total intensity remains nearly constant. This observation is not encompassed within the framework of current theoretical models of two-photon lasers and results from the atomic Zeeman degeneracy of the states participating in the two-photon stimulated emission process. We suggest that such a two-photon laser could emit polarization-entangled twin beams of light if this degeneracy is lifted, which may have quantum information applications.

To achieve two-photon optical lasing, we use a two-photon gain medium consists of a dense, collimated, effusive beam of laser-driven ^{39}K atoms that passes through the center of a high-finesse optical resonator. We use the D_1 transition of ^{39}K whose atomic states are denoted by $|\alpha F_\alpha M_\alpha\rangle$, with $\alpha = g$ for $4^2S_{1/2}$ and $\alpha = e$ for $4^2P_{1/2}$ levels and with a ground-state hyperfine splitting of $\Delta_g = 462$ MHz. To create a population inversion for multiphoton transitions starting from the state $|g22\rangle$, the atoms are continuously optically pumped into this state by two $\hat{\sigma}_+$ -polarized fields whose frequencies are set close to ω_{21} and ω_{22} (Bohr frequencies are denoted by $\omega_{F_e F_g}$). The two-photon laser resonator is in a linear (standing-wave) transverse-mode-degenerate sub-confocal configuration with a cavity finesse of 15,140 and contains a maximum of 7×10^6 atoms.

Two-photon stimulated emission into the laser cavity mode (frequency ω_l) occurs when the optically pumped atoms are illuminated by an intense $\hat{\sigma}_-$ -polarized Raman pump beam of frequency ω_d , and ω_l is adjusted to the two-photon transition frequency ($\simeq \omega_d + \Delta_g/2$). Photons are added two at a time to the cavity field via stimulated emission when two photons are annihilated from the $\hat{\sigma}_-$ -polarized Raman pump field and the atom undergoes a multi-photon transition from the state $|g22\rangle$ to any one of the states $|g1M_g\rangle$. Note that such a hyper-Raman process can be mapped onto the idealized two-photon stimulated emission process by working in a dressed-state basis. The atomic, Raman pump beam, and optical cavity axes are mutually orthogonal, thereby suppressing competing phase-matched processes that might give rise to parametric oscillation.

Conclusive evidence that we have achieved two-photon lasing is indicated by the threshold behavior of the laser. Due to the nonlinear gain, lasing will only commence when a critical atomic inversion density and a critical cavity photon number is present. To observe the threshold behavior, we measure the power of the beam emitted from

the two-photon laser resonator and inject pulses of light into the cavity to trigger the laser to the 'on' state and shown in Fig. 1(a) for $\sim 3.3 \times 10^5$ injected photons. The duration of the injected pulse is indicated by the dashed vertical lines. It is seen that the power emitted by the cavity rises to approximately $0.2 \mu\text{W}$ and remains at this value, corresponding to an average cavity photon number of 2.2×10^6 and an intracavity intensity of 7.3 W/cm^2 at the cavity waist.

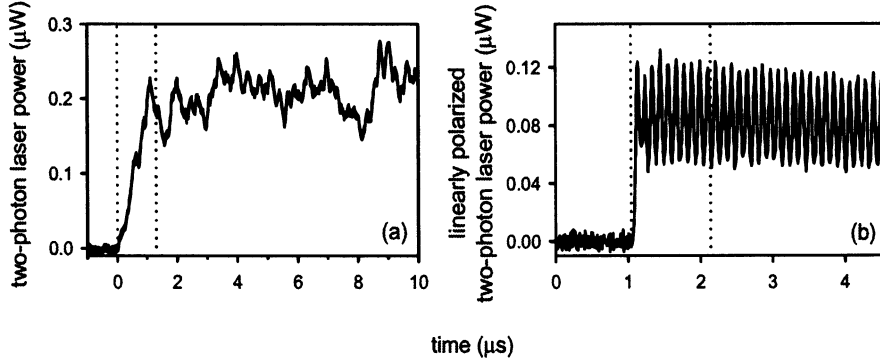


Fig. 1. (a) Turn on behavior of the two-photon laser. (b) Polarization dynamics of the two-photon laser.

While the total emitted power is seen to remain nearly constant, we find that the two-photon laser displays polarization instabilities. By placing a linear polarization before the detector, it is seen in Fig. 1(b) that the state of polarization undergoes very regular oscillations of period $0.11 \mu\text{s}$, with a 50% depth of modulation. Similar dynamical behavior is observed for all polarizer orientations, suggesting that the state of polarization is elliptical with an ellipticity of 0.5 and a rotating major axis. We find that the laser dynamics is quite sensitive to the applied magnetic field: an increase by as little as 0.5 Gauss is sufficient to generate a much more complicated pattern, turning into a possibly chaotic pattern for a field strength of 2 Gauss. We believe that it may be possible to suppress the instabilities using a stronger magnetic field to lift the degeneracy of the different quantum pathways.

Lifting the degeneracy between the pathways also has the promise of yielding polarization entanglement. Consider, for simplicity, a single emission event. When the Zeeman levels are degenerate, the superposition of all possible amplitudes gives a photon pair (1,2) in the state $[\alpha_1|\hat{z}\rangle_1 + \beta_1|\hat{x}\rangle_1][\alpha_2|\hat{z}\rangle_2 + \beta_2|\hat{x}\rangle_2]$ with $|\alpha_i|^2 + |\beta_i|^2 = 1$, where $|\hat{\epsilon}\rangle_i$ ($\epsilon = x, y$) is the polarization-labeled Fock state of the i^{th} photon of the emitted pair. In a strong magnetic field, the $\hat{x}\hat{z}$ and $\hat{z}\hat{x}$ pairs become off-resonant and should stop lasing, creating the polarization-entangled state $\alpha_1\alpha_2|\hat{z}\rangle_1|\hat{z}\rangle_2 + \beta_1\beta_2|\hat{x}\rangle_1|\hat{x}\rangle_2$.

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References and links

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